Lecture Notes on Optimization and Optimal Control

Module 6: Algorithms for Unconstrained Problems

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This module develops a unified and rigorous treatment of modern optimization methods for unconstrained problems. Beginning with first-order methods, the notes analyze gradient descent under exact and perturbed gradients, emphasizing stability, descent properties, and convergence under deterministic and martingale noise. Stochastic gradient and Robbins–Monro algorithms are formulated within the general stochastic approximation framework, and their convergence is established using Lyapunov techniques and the ODE method.

The exposition then turns to second-order methods. Newton's method is derived from local quadratic modeling, and its global and local convergence behavior is analyzed in detail. Under strong convexity and Lipschitz Hessian assumptions, the method exhibits a damped global descent phase followed by a local quadratic convergence regime. The Newton decrement is interpreted as a natural measure of optimality and as an accurate predictor of local objective decrease.

Throughout, the notes emphasize a dynamical-systems viewpoint: optimization algorithms are interpreted as discrete-time approximations of continuous flows. This perspective unifies deterministic descent, stochastic approximation, fixed-point iterations, and second-order methods within a common analytical framework based on Lyapunov stability, contraction principles, and martingale convergence theory.

1 Numerical Algorithms for Unconstrained Optimization

We consider the unconstrained optimization problem

$$\min_{r \in \mathbb{R}^n} f(r), \tag{1.1}$$

where the objective function $f : \mathbb{R}^n \to \mathbb{R}$ is assumed to be continuously differentiable.

Unconstrained smooth optimization constitutes one of the central paradigms of optimization theory and arises throughout engineering, control, economics, and machine learning. Except in special cases such as quadratic objectives or separable structures, problem (1.1) does not admit a closed-form solution. Consequently, one relies on numerical algorithms that iteratively generate approximations to minimizers.

1.1 Iterative Descent Framework

A cornerstone of smooth optimization is the first-order necessary condition for local optimality: if $r^\star$ is a local minimizer of f, then

$$\nabla f(r^\star) = 0.$$

Points satisfying this condition are called *stationary points*. In the absence of convexity, a stationary point may correspond to a local minimum, a local maximum, or a saddle point. Thus, stationarity is a necessary but not sufficient condition for optimality in general. Nevertheless, most first-order numerical algorithms are constructed so that any limit point of the generated sequence, when it exists, satisfies the stationarity condition.

Since problem (1.1) rarely admits a closed-form solution, numerical methods generate a sequence

$$\{r_t\}_{t \geq 0} \subset \mathbb{R}^n$$

intended to approach a stationary point of f. A broad and unifying representation of such methods is the iteration

$$r_{t+1} = r_t + \gamma_t s_t, \qquad t = 0, 1, 2, \ldots, \tag{1.2}$$

where $s_t \in \mathbb{R}^n$ is a search direction and $\gamma_t > 0$ is a step size.

This formulation separates the algorithmic design into two fundamental components: the choice of direction s_t, which determines how local information about the objective is used, and the step-size rule γ_t, which controls stability and progress along that direction. The guiding

principle of descent methods is that each iteration produces a sufficient decrease in the objective value, thereby driving the sequence toward a stationary point.

1.2 Descent Directions

The essential structural requirement for a descent-based method is that the search direction yields a local decrease of the objective function.

Definition 1.1 (Descent direction)**.** A vector $s \in \mathbb{R}^n$ is called a *descent direction* for f at a point r if

$$\nabla f(r)^\top s < 0.$$

This condition admits a clear geometric interpretation. The gradient $\nabla f(r)$ identifies the direction of steepest local increase of f under the Euclidean norm. The directional derivative of f at r along s is

$$Df(r)[s] = \nabla f(r)^\top s.$$

Hence, the inequality $\nabla f(r)^\top s < 0$ means that f decreases locally in the direction s.

The descent property follows directly from the first-order Taylor expansion:

$$f(r + \gamma s) = f(r) + \gamma \nabla f(r)^\top s + o(\gamma), \qquad \gamma \to 0.$$

If $\nabla f(r)^\top s < 0$, then there exists $\bar{\gamma} > 0$ such that

$$f(r + \gamma s) < f(r), \qquad \forall \gamma \in (0, \bar{\gamma}).$$

Thus, descent directions guarantee strict local decrease for sufficiently small step sizes.

If $\nabla f(r) = 0$, then $\nabla f(r)^\top s = 0$ for all $s \in \mathbb{R}^n$, so no descent direction exists. This explains why first-order methods naturally terminate at stationary points.

1.3 Choice of Descent Direction

Different numerical methods arise from different choices of the search direction s_t, reflecting how much local information about the objective function is exploited.

1.3.1 Steepest Descent Method

The steepest descent (or gradient) method selects

$$s_t = -\nabla f(r_t). \tag{1.3}$$

The gradient points in the direction of steepest local increase of the objective; therefore its negative corresponds to the direction of steepest local decrease under the Euclidean norm.

Among all unit-norm directions d, the direction minimizing the directional derivative $\nabla f(r_t)^\top d$ is

$$d = -\frac{\nabla f(r_t)}{\|\nabla f(r_t)\|}.$$

Thus, the steepest descent direction is locally optimal in a first-order sense. Despite its simplicity, the method may converge slowly when the level sets of f are highly elongated, i.e., when the problem is ill-conditioned.

1.3.2 Newton's Method

Newton's method incorporates second-order information through the Hessian matrix. The search direction is defined as

$$s_t = -\left(\nabla^2 f(r_t)\right)^{-1} \nabla f(r_t), \tag{1.4}$$

assuming that the Hessian $\nabla^2 f(r_t)$ is symmetric and positive definite.

Quadratic model interpretation. Newton's method can be derived by minimizing the second-order Taylor approximation of f at r_t,

$$\tilde{f}_t(r) = f(r_t) + \nabla f(r_t)^\top (r - r_t) + \frac{1}{2}(r - r_t)^\top \nabla^2 f(r_t)(r - r_t). \tag{1.5}$$

Setting the gradient of this quadratic model equal to zero yields

$$\nabla f(r_t) + \nabla^2 f(r_t)(r - r_t) = 0,$$

which leads to the update

$$r_{t+1} = r_t - \left(\nabla^2 f(r_t)\right)^{-1} \nabla f(r_t). \tag{1.6}$$

When f is twice continuously differentiable and strongly convex in a neighborhood of a minimizer, Newton's method exhibits quadratic local convergence. However, if the Hessian is

indefinite or poorly conditioned, the method may fail or require damping or line-search strategies to ensure global convergence.

1.3.3 Quasi-Newton Methods

Quasi-Newton methods approximate the Newton direction without explicitly computing or inverting the Hessian. The search direction takes the form

$$s_t = -D_t \nabla f(r_t), \tag{1.7}$$

where D_t is a symmetric positive definite matrix intended to approximate $(\nabla^2 f(r_t))^{-1}$.

The matrix D_t is updated iteratively using gradient differences so as to satisfy secant conditions that capture curvature information. Well-known quasi-Newton schemes include BFGS and limited-memory BFGS (L-BFGS). Under suitable assumptions, these methods achieve superlinear local convergence while maintaining moderate computational cost.

1.4 Step Size Rules

The scalar $\gamma_t > 0$ in the update

$$r_{t+1} = r_t + \gamma_t s_t$$

is called the *step size* (or learning rate). Its selection plays a decisive role in determining both convergence and efficiency.

Constant Step Size. A constant step size $\gamma_t = \gamma$ is simple to implement. If γ is too large, the iterates may oscillate or diverge; if too small, convergence may be excessively slow. Constant step sizes are therefore most effective when the problem is well conditioned or when local convergence is of primary interest.

Exact and Bounded Line Search. An alternative is to select γ_t by minimizing the objective along the search direction, $\gamma_t = \arg\min_{\gamma \geq 0} f(r_t + \gamma s_t)$. This is known as an exact line search. In practice, one often restricts the search to a bounded interval $\gamma \in [0, \bar{\gamma}]$, leading to bounded line-search methods. Although effective for classical smooth problems, such procedures may be computationally burdensome in large-scale settings.

Diminishing Step Size. A third important class of rules consists of diminishing step sizes satisfying

$$\gamma_t \to 0, \qquad \sum_{t=0}^{\infty} \gamma_t = \infty. \tag{1.8}$$

The first condition ensures increasingly conservative steps, while the second prevents premature stagnation.

To see why these conditions exclude convergence to nonstationary points, suppose that $r_t \to \bar{r}$ with $\nabla f(\bar{r}) \neq 0$. For sufficiently large indices $m > n$, we have

$$r_m \approx r_n \approx \bar{r},$$

and therefore

$$r_m \approx r_n - \left(\sum_{t=n}^{m-1} \gamma_t \right) \nabla f(\bar{r}).$$

Since $\sum_{t=n}^{m-1} \gamma_t$ can be made arbitrarily large by (1.8), this contradicts convergence. Hence, any accumulation point must be stationary.

2 Convergence Results

We now investigate the convergence properties of the general descent iteration

$$r_{t+1} = r_t + \gamma_t s_t, \tag{2.1}$$

where s_t is a search direction and $\gamma_t > 0$ is a step size.

A first-order Taylor expansion of f around r_t yields

$$f(r_{t+1}) = f(r_t + \gamma_t s_t) = f(r_t) + \gamma_t \nabla f(r_t)^\top s_t + o(\gamma_t), \qquad \gamma_t \to 0,$$

where $o(\gamma_t)$ denotes higher-order terms satisfying $o(\gamma_t)/\gamma_t \to 0$.

This expansion makes explicit the local mechanism governing descent: the term $\nabla f(r_t)^\top s_t$ determines whether the objective decreases to first order, while the higher-order terms must be controlled through regularity assumptions.

Fundamental Question. Under what conditions does every limit point of the sequence generated by (2.1) correspond to a stationary point of f?

The Taylor expansion alone does not guarantee convergence to stationarity. A potential failure mode arises if

$$\nabla f(r_t)^\top s_t \to 0 \quad \text{while} \quad \nabla f(r_t) \not\to 0,$$

that is, if the search direction becomes asymptotically orthogonal to the gradient. In such a situation, the algorithm may cease to make progress even though the gradient remains nonzero, leading to stagnation at a nonstationary point.

To exclude this pathological behavior, one imposes quantitative conditions ensuring that the search direction maintains a uniformly favorable angle with respect to the negative gradient and does not grow excessively in magnitude.

Assumption 2.1 (Uniform descent conditions)**.** There exist constants $C_1 > 0$ and $C_2 > 0$ such that, for all t,

$$C_1 \|\nabla f(r_t)\|^2 \leq -\nabla f(r_t)^\top s_t, \tag{2.2}$$

$$\|s_t\| \leq C_2 \|\nabla f(r_t)\|. \tag{2.3}$$

Condition (2.2) ensures a uniform amount of descent proportional to the squared gradient norm. It rules out directions that become nearly orthogonal to the gradient and guarantees that the decrease in the objective is quantitatively significant whenever the gradient is nonzero.

Condition (2.3) controls the size of the search direction relative to the gradient. This prevents the step $\gamma_t s_t$ from being dominated by excessively large directions that could invalidate the local Taylor approximation or destabilize the iteration.

Together, these conditions ensure that the descent mechanism remains effective and well scaled throughout the iterative process.

For the steepest descent method, where $s_t = -\nabla f(r_t)$, both inequalities in Assumption 2.1 hold trivially with $C_1 = C_2 = 1$. Under suitable regularity assumptions—such as positive definiteness and boundedness of the Hessian—Newton and quasi-Newton directions also satisfy these conditions, thereby fitting into the same convergence framework.

2.1 Convergence for a Constant Step Size

We analyze the convergence of the iteration

$$r_{t+1} = r_t + \gamma s_t, \tag{2.4}$$

with a constant step size $\gamma > 0$.

Assumption 2.2 (Lipschitz continuity of the gradient)**.** There exists a constant $L > 0$ such that

$$\|\nabla f(r) - \nabla f(\bar{r})\| \leq L\|r - \bar{r}\|, \qquad \forall r, \bar{r} \in \mathbb{R}^n.$$

Assumption 2.3 (Uniform descent conditions)**.** There exist constants $C_1, C_2 > 0$ such that, for all t,

$$C_1\|\nabla f(r_t)\|^2 \leq -\nabla f(r_t)^\top s_t, \qquad \|s_t\| \leq C_2\|\nabla f(r_t)\|.$$

Lemma 2.4 (Quadratic upper bound under Lipschitz gradient)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and suppose its gradient is L-Lipschitz, i.e.,*

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \qquad \forall x, y \in \mathbb{R}^n.$$

Then, for all $r, z \in \mathbb{R}^n$,

$$f(r + z) \leq f(r) + z^\top \nabla f(r) + \frac{L}{2}\|z\|^2. \tag{2.5}$$

Proof. Fix $r, z \in \mathbb{R}^n$ and define the scalar function

$$g(\xi) := f(r + \xi z), \qquad \xi \in [0, 1].$$

Since f is continuously differentiable, the chain rule gives

$$g'(\xi) = \nabla f(r + \xi z)^\top z.$$

By the fundamental theorem of calculus,

$$f(r + z) - f(r) = g(1) - g(0) = \int_0^1 g'(\xi)\, d\xi = \int_0^1 z^\top \nabla f(r + \xi z)\, d\xi. \tag{2.6}$$

We now add and subtract $\nabla f(r)$ inside the integrand:

$$\begin{aligned} f(r + z) - f(r) &= \int_0^1 z^\top \nabla f(r)\, d\xi + \int_0^1 z^\top (\nabla f(r + \xi z) - \nabla f(r)) d\xi \\ &= z^\top \nabla f(r) + \int_0^1 z^\top (\nabla f(r + \xi z) - \nabla f(r)) d\xi. \end{aligned} \tag{2.7}$$

To estimate the second term, apply the Cauchy–Schwarz inequality:

$$z^\top(\nabla f(r+\xi z) - \nabla f(r)) \le \|z\|\,\|\nabla f(r+\xi z) - \nabla f(r)\|.$$

By the Lipschitz continuity of the gradient,

$$\|\nabla f(r+\xi z) - \nabla f(r)\| \le L\|r+\xi z - r\| = L\xi\|z\|.$$

Therefore,

$$z^\top(\nabla f(r+\xi z) - \nabla f(r)) \le L\xi\|z\|^2.$$

Substituting this estimate into (2.7) gives

$$\begin{aligned} f(r+z) - f(r) &\le z^\top \nabla f(r) + \int_0^1 L\xi\|z\|^2\,d\xi \\ &= z^\top \nabla f(r) + L\|z\|^2 \int_0^1 \xi\,d\xi. \end{aligned}$$

Since $\int_0^1 \xi\,d\xi = \frac{1}{2}$, we obtain

$$f(r+z) - f(r) \le z^\top \nabla f(r) + \frac{L}{2}\|z\|^2,$$

which establishes (2.5). □

Proposition 2.5 (Convergence with constant step size)**.** *Let $\{r_t\}$ be generated by* (2.4) *and suppose Assumptions 2.2 and 2.3 hold. If*

$$0 < \gamma < \frac{2C_1}{LC_2^2}, \tag{2.8}$$

then either $f(r_t) \to -\infty$, or $f(r_t)$ converges to a finite value and

$$\lim_{t\to\infty} \nabla f(r_t) = 0.$$

Moreover, every limit point of $\{r_t\}$ is a stationary point of f.

Proof. Applying Lemma 2.4 with $r = r_t$ and $z = \gamma s_t$, we obtain

$$f(r_{t+1}) - f(r_t) \le \gamma \nabla f(r_t)^\top s_t + \frac{L}{2}\gamma^2\|s_t\|^2. \tag{2.9}$$

Using the descent conditions in Assumption 2.3,

$$-\nabla f(r_t)^\top s_t \geq C_1 \|\nabla f(r_t)\|^2, \qquad \|s_t\|^2 \leq C_2^2 \|\nabla f(r_t)\|^2.$$

Substituting these bounds into (2.9) yields

$$\begin{aligned} f(r_{t+1}) - f(r_t) &\leq -\gamma C_1 \|\nabla f(r_t)\|^2 + \frac{L}{2}\gamma^2 C_2^2 \|\nabla f(r_t)\|^2 \\ &= -\gamma \left(C_1 - \frac{L}{2}\gamma C_2^2 \right) \|\nabla f(r_t)\|^2. \end{aligned} \tag{2.10}$$

Equivalently,

$$f(r_t) - f(r_{t+1}) \geq \gamma \left(C_1 - \frac{L}{2}\gamma C_2^2 \right) \|\nabla f(r_t)\|^2. \tag{2.11}$$

Condition (2.8) guarantees that

$$\alpha := \gamma \left(C_1 - \frac{L}{2}\gamma C_2^2 \right) > 0.$$

Hence

$$f(r_{t+1}) \leq f(r_t) - \alpha \|\nabla f(r_t)\|^2, \tag{2.12}$$

which shows that $\{f(r_t)\}$ is nonincreasing.

If $f(r_t) \to -\infty$, the conclusion holds. Suppose instead that $f(r_t)$ is bounded below. Then, since it is monotone, $f(r_t)$ converges to a finite limit.

Summing (2.11) from $t = 0$ to $T - 1$ gives

$$\sum_{t=0}^{T-1} \alpha \|\nabla f(r_t)\|^2 \leq f(r_0) - f(r_T).$$

Letting $T \to \infty$ and using boundedness of $f(r_T)$, we obtain

$$\sum_{t=0}^{\infty} \|\nabla f(r_t)\|^2 < \infty.$$

Therefore,

$$\lim_{t \to \infty} \|\nabla f(r_t)\| = 0.$$

Finally, let $r_{t_k} \to \bar{r}$ be a convergent subsequence. Since $\nabla f(r_{t_k}) \to 0$ and ∇f is continuous,

$$\nabla f(\bar{r}) = \lim_{k \to \infty} \nabla f(r_{t_k}) = 0.$$

Hence every limit point $\bar{r}$ is stationary, completing the proof. $\square$

2.2 Convergence for a Diminishing Step Size

We now consider the iteration

$$r_{t+1} = r_t + \gamma_t s_t, \tag{2.13}$$

with diminishing step sizes.

Proposition 2.6 (Convergence with diminishing step size)**.** *Suppose Assumptions 2.2 and 2.3 hold, and that*

$$\gamma_t \to 0, \qquad \sum_{t=0}^{\infty} \gamma_t = \infty. \tag{2.14}$$

Then either $f(r_t) \to -\infty$, or $f(r_t)$ converges to a finite value and

$$\lim_{t\to\infty} \nabla f(r_t) = 0.$$

Every limit point of $\{r_t\}$ is a stationary point of f.

Proof. Applying the quadratic upper bound (2.5) with $r = r_t$ and $z = \gamma_t s_t$ yields

$$f(r_{t+1}) - f(r_t) \le \gamma_t \nabla f(r_t)^\top s_t + \frac{L}{2}\gamma_t^2 \|s_t\|^2.$$

Using the descent conditions

$$-\nabla f(r_t)^\top s_t \ge C_1 \|\nabla f(r_t)\|^2, \qquad \|s_t\|^2 \le C_2^2 \|\nabla f(r_t)\|^2,$$

we obtain

$$\begin{aligned} f(r_{t+1}) - f(r_t) &\le -\gamma_t C_1 \|\nabla f(r_t)\|^2 + \frac{L}{2}\gamma_t^2 C_2^2 \|\nabla f(r_t)\|^2 \\ &= -\gamma_t \left(C_1 - \frac{L}{2}\gamma_t C_2^2 \right) \|\nabla f(r_t)\|^2. \end{aligned} \tag{2.15}$$

Since $\gamma_t \to 0$, there exists $\bar{t}$ such that for all $t \ge \bar{t}$,

$$\gamma_t \le \frac{C_1}{LC_2^2}.$$

Hence, for $t \ge \bar{t}$,

$$C_1 - \frac{L}{2}\gamma_t C_2^2 \ge \frac{C_1}{2}.$$

Substituting into (2.15) gives

$$f(r_{t+1}) \leq f(r_t) - \frac{C_1}{2}\gamma_t \|\nabla f(r_t)\|^2, \qquad t \geq \bar{t}. \tag{2.16}$$

Thus, for sufficiently large t, the objective decreases by an amount proportional to $\gamma_t \|\nabla f(r_t)\|^2$. In particular, $\{f(r_t)\}$ is eventually nonincreasing.

If $f(r_t) \to -\infty$, the conclusion holds. Assume instead that $f(r_t)$ is bounded below. Since it is eventually monotone decreasing, it converges to a finite limit.

Summing (2.16) from $t = \bar{t}$ to $T - 1$ yields

$$\frac{C_1}{2} \sum_{t=\bar{t}}^{T-1} \gamma_t \|\nabla f(r_t)\|^2 \leq f(r_{\bar{t}}) - f(r_T).$$

Letting $T \to \infty$ and using boundedness of $f(r_T)$, we obtain

$$\sum_{t=\bar{t}}^{\infty} \gamma_t \|\nabla f(r_t)\|^2 < \infty.$$

Suppose, by contradiction, that $\|\nabla f(r_t)\| \not\to 0$. Then there exist $\varepsilon > 0$ and an infinite subsequence $\{t_k\}$ such that

$$\|\nabla f(r_{t_k})\| \geq \varepsilon.$$

Since $\gamma_t > 0$ and $\sum_{t=0}^{\infty} \gamma_t = \infty$, the sum

$$\sum_k \gamma_{t_k}$$

must diverge. Therefore,

$$\sum_k \gamma_{t_k} \|\nabla f(r_{t_k})\|^2 \geq \varepsilon^2 \sum_k \gamma_{t_k} = \infty,$$

which contradicts the finiteness of $\sum_{t=\bar{t}}^{\infty} \gamma_t \|\nabla f(r_t)\|^2$. Hence,

$$\lim_{t\to\infty} \|\nabla f(r_t)\| = 0.$$

Finally, let $r_{t_k} \to \bar{r}$ along a convergent subsequence. By continuity of ∇f,

$$\nabla f(\bar{r}) = \lim_{k\to\infty} \nabla f(r_{t_k}) = 0,$$

so every limit point is stationary. □

2.3 Example: Steepest Descent on a Quadratic Function

We now present a fully formal analysis of the steepest descent method applied to a quadratic objective. This example serves as a canonical model in smooth optimization, since the iteration becomes linear and all convergence properties can be characterized exactly.

Gradient iteration. Consider the gradient descent iteration

$$r_{t+1} = r_t - \gamma_t \nabla f(r_t), \qquad t = 0, 1, 2, \ldots, \tag{2.17}$$

where $\gamma_t > 0$ is the step size.

Rates of convergence. Let $r^\star$ be a stationary point of f. The sequence $\{r_t\}$ is said to converge:

1. *Linearly* (or geometrically) if there exist constants $\rho \in (0, 1)$ and $C > 0$ such that

$$\|r_t - r^\star\| \leq C\rho^t, \qquad \forall t \geq 0. \tag{2.18}$$

2. *Sublinearly* if $\|r_t - r^\star\| \to 0$ but no geometric rate holds, for example,

$$\|r_t - r^\star\| = \mathcal{O}\left(\frac{1}{t}\right). \tag{2.19}$$

3. In *finite time* if there exists $T < \infty$ such that $r_t = r^\star$ for all $t \geq T$.

Equivalent notions can be formulated in terms of objective values $f(r_t) - f(r^\star)$ or gradient norms $\|\nabla f(r_t)\|$.

Quadratic objective. Consider the quadratic function

$$f(r) = \frac{1}{2} r^\top Q r, \tag{2.20}$$

where $Q \in \mathbb{R}^{n \times n}$ is symmetric positive definite. Then f is strongly convex, the unique minimizer is

$$r^\star = 0,$$

and the gradient satisfies

$$\nabla f(r) = Qr.$$

Substituting into (2.17) gives

$$r_{t+1} = (I - \gamma_t Q) r_t. \tag{2.21}$$

Thus the method reduces to a linear time-varying dynamical system.

Constant step size

Assume $\gamma_t = \gamma$ is constant. Then (2.21) yields

$$r_t = (I - \gamma Q)^t r_0. \tag{2.22}$$

Let $\{\lambda_i\}_{i=1}^n$ denote the eigenvalues of Q. Since $Q \succ 0$, we have

$$0 < \lambda_{\min}(Q) \leq \lambda_i \leq \lambda_{\max}(Q).$$

The eigenvalues of $I - \gamma Q$ are $1 - \gamma\lambda_i$, and the spectral radius satisfies

$$\rho(I - \gamma Q) = \max_{1 \leq i \leq n} |1 - \gamma\lambda_i|.$$

Proposition 2.7. *The iteration* (2.22) *converges to zero for all initial conditions* r_0 *if and only if*

$$0 < \gamma < \frac{2}{\lambda_{\max}(Q)}. \tag{2.23}$$

In this case,

$$\|r_t\| \leq \rho^t \|r_0\|, \qquad \rho = \rho(I - \gamma Q) < 1,$$

and convergence is linear.

Proof. Convergence holds if and only if $\rho(I - \gamma Q) < 1$. Since

$$|1 - \gamma\lambda_i| < 1 \quad \forall i,$$

this is equivalent to

$$-1 < 1 - \gamma\lambda_i < 1, \quad \forall i.$$

The right inequality gives $\gamma\lambda_i > 0$, which holds for $\gamma > 0$. The left inequality yields $\gamma\lambda_i < 2$ for all i, hence $\gamma < 2/\lambda_{\max}(Q)$. This proves (2.23). □

Optimal constant step size and conditioning. The contraction factor is minimized by choosing

$$\gamma^\star = \frac{2}{\lambda_{\min}(Q) + \lambda_{\max}(Q)},$$

which yields

$$\rho^\star = \frac{\kappa(Q) - 1}{\kappa(Q) + 1}, \qquad \kappa(Q) = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}.$$

As $\kappa(Q) \to \infty$, we have $\rho^\star \to 1$, and convergence becomes arbitrarily slow. Thus the linear rate depends explicitly on the condition number of the Hessian.

Diminishing step sizes

If $\gamma_t \to 0$, for example $\gamma_t = c/t$, the iteration (2.21) becomes time-varying. In this regime the spectral radius argument no longer yields a uniform contraction factor. Instead one typically obtains sublinear convergence rates, such as

$$f(r_t) - f(r^\star) = \mathcal{O}\left(\frac{1}{t}\right), \qquad \|\nabla f(r_t)\| = \mathcal{O}\left(\frac{1}{\sqrt{t}}\right).$$

These rates coincide with those established in the general smooth optimization framework without strong convexity.

Remark 2.8. The quadratic objective provides an exact realization of the general theory. Lipschitz continuity of the gradient corresponds to $\|\nabla^2 f\| \le L$, while strong convexity corresponds to $\nabla^2 f \succeq mI$. In this setting the quadratic upper and lower bounds become exact, and the distinction between linear and sublinear convergence is completely characterized by the spectral properties of Q.

2.4 Strong Convexity and Gradient Descent

We now state and prove a precise convergence theorem for gradient descent under smoothness and strong convexity assumptions.

Definition 2.9 (Strong convexity and smoothness)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable.

1. The function f is *strongly convex with parameter* $m > 0$ if

$$\nabla^2 f(x) \succeq mI, \qquad \forall x \in \mathbb{R}^n. \tag{2.24}$$

2. The function f is *L-smooth* if its gradient is L-Lipschitz continuous, that is,

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \qquad \forall x, y \in \mathbb{R}^n. \tag{2.25}$$

Strong convexity is equivalent to the quadratic lower bound

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2}\|y - x\|^2, \qquad \forall x, y \in \mathbb{R}^n. \tag{2.26}$$

Under assumptions (2.24)–(2.25), the function f admits a unique minimizer $x^\star$, characterized by

$$\nabla f(x^\star) = 0.$$

Theorem 2.10 (Linear convergence of gradient descent)**.** *Let f be m-strongly convex and L-smooth. Consider the gradient descent iteration*

$$x_{k+1} = x_k - t\nabla f(x_k), \tag{2.27}$$

with constant step size $t > 0$. If

$$0 < t < \frac{2m}{L^2}, \tag{2.28}$$

then the iterates converge linearly to the unique minimizer $x^\star$. More precisely,

$$\|x_k - x^\star\| \leq \rho^k \|x_0 - x^\star\|, \tag{2.29}$$

where

$$\rho = \sqrt{1 - 2mt + t^2L^2} \in (0, 1).$$

Proof. Subtracting $x^\star$ from both sides of (2.27) gives

$$x_{k+1} - x^\star = x_k - x^\star - t\big(\nabla f(x_k) - \nabla f(x^\star)\big).$$

Taking squared norms yields

$$\begin{aligned} \|x_{k+1} - x^\star\|^2 = \|x_k - x^\star\|^2 + t^2\|\nabla f(x_k) - \nabla f(x^\star)\|^2 \\ - 2t(x_k - x^\star)^\top \big(\nabla f(x_k) - \nabla f(x^\star)\big). \end{aligned} \tag{2.30}$$

By L-smoothness,

$$\|\nabla f(x_k) - \nabla f(x^\star)\| \leq L\|x_k - x^\star\|.$$

By strong convexity,

$$(\nabla f(x_k) - \nabla f(x^\star))^\top (x_k - x^\star) \geq m\|x_k - x^\star\|^2.$$

Substituting these bounds into (2.30) yields

$$\|x_{k+1} - x^\star\|^2 \leq \left(1 - 2mt + t^2L^2\right) \|x_k - x^\star\|^2.$$

Define

$$\theta(t) = 1 - 2mt + t^2L^2.$$

Condition (2.28) guarantees that $\theta(t) \in (0,1)$. Let $\rho = \sqrt{\theta(t)} \in (0,1)$. Then

$$\|x_{k+1} - x^\star\| \leq \rho\|x_k - x^\star\|.$$

Iterating this inequality proves (2.29). □

Proposition 2.11 (Optimal step size and conditioning)**.** *Under the assumptions of Theorem 2.10, the contraction factor*

$$\theta(t) = 1 - 2mt + t^2L^2$$

is minimized at

$$t^\star = \frac{m}{L^2}.$$

In the quadratic case $f(x) = \frac{1}{2}x^\top Qx$, one has $m = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$. Hence the convergence rate depends explicitly on the ratio

$$\kappa(Q) = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)},$$

which is the condition number of Q. As $\kappa(Q)$ increases, the contraction factor approaches one, and convergence becomes arbitrarily slow.

3 Gradient Methods with Noise or Errors

In many practical optimization problems, the gradient $\nabla f(r_t)$ is not available in exact form. Instead, the algorithm has access to an approximation that can be written as

$$\nabla f(r_t) + w_t,$$

where $w_t \in \mathbb{R}^n$ denotes an error or noise term. Such perturbations arise naturally in a variety of settings. In deterministic numerical computation, round-off errors and finite-precision arithmetic introduce discrepancies between the true and computed gradient. In problems involving discretization, such as PDE-constrained optimization, numerical approximation of derivatives generates additional errors. In data-driven and large-scale learning problems, gradients are often estimated from samples, leading to stochastic fluctuations due to sampling variability.

When the gradient is perturbed in this way, the update rule becomes

$$r_{t+1} = r_t - \gamma_t\big(\nabla f(r_t) + w_t\big), \tag{3.1}$$

where $\gamma_t > 0$ is the step size. The iteration (3.1) is commonly referred to as a *perturbed gradient method* or, in stochastic settings, *stochastic gradient descent*.

Without additional assumptions on the noise sequence $\{w_t\}$, convergence cannot, in general, be guaranteed. If the perturbations are arbitrary, adversarial, or unbounded, the iterates may diverge or fail to approach a stationary point. A meaningful convergence analysis therefore requires structural assumptions on the noise, such as boundedness, vanishing magnitude, martingale-difference structure, or bounded variance. These structured noise models reflect the mechanisms generating the perturbations and allow one to establish convergence guarantees under appropriate step-size conditions.

3.1 Representative Noise Models

We now describe several representative perturbation models that arise naturally in deterministic and stochastic optimization and illustrate their qualitative effect on convergence.

1. Error small relative to the gradient Suppose that the perturbation satisfies

$$\|w_t\| \le \varepsilon_t \|\nabla f(r_t)\|, \qquad 0 \le \varepsilon_t < 1,$$

for all t. Consider the perturbed search direction

$$s_t = -\nabla f(r_t) - w_t.$$

Then

$$\begin{aligned}
\nabla f(r_t)^\top s_t &= -\|\nabla f(r_t)\|^2 - \nabla f(r_t)^\top w_t \\
&\leq -\|\nabla f(r_t)\|^2 + \|\nabla f(r_t)\| \, \|w_t\| \\
&\leq -(1-\varepsilon_t)\|\nabla f(r_t)\|^2.
\end{aligned}$$

Hence, whenever $\varepsilon_t < 1$ and $\nabla f(r_t) \neq 0$, the direction s_t remains a descent direction. In particular, relative errors that are uniformly bounded away from one do not destroy the descent property, and the qualitative behavior of gradient descent is preserved.

2. Uniformly bounded error Assume that the perturbation is uniformly bounded:

$$\|w_t\| \leq \delta, \qquad \forall t,$$

for some constant $\delta > 0$.

If the noise is time-invariant, that is, $w_t = w$ for all t, then the update

$$r_{t+1} = r_t - \gamma_t\big(\nabla f(r_t) + w\big)$$

is equivalent to applying gradient descent to the perturbed objective

$$\tilde{f}(r) = f(r) + w^\top r.$$

A stationary point $\bar{r}$ of $\tilde{f}$ satisfies

$$\nabla f(\bar{r}) = -w.$$

Thus the method converges to the minimizer of a linearly perturbed objective rather than to a true stationary point of f.

If w_t varies with time but remains bounded, exact convergence to a stationary point of f cannot generally be expected. Instead, the iterates typically enter and remain within a neigh-

borhood of the stationary set characterized by

$$\|\nabla f(r)\| \leq \delta,$$

so that the residual gradient is controlled by the noise magnitude.

3. Step-size–proportional error A particularly important and benign noise model assumes that

$$\|w_t\| \leq \gamma_t\big(q + p\|\nabla f(r_t)\|\big), \qquad \forall t, \tag{3.2}$$

for some constants $p, q \geq 0$.

In this case the perturbation vanishes asymptotically whenever $\gamma_t \to 0$. Such errors naturally arise in stochastic approximation, incremental gradient methods, and online learning algorithms. Because the noise magnitude scales with the step size, the cumulative perturbation remains controlled, and under standard diminishing step-size conditions one can recover convergence to a stationary point of f.

3.2 General Convergence Result with Errors

We now consider a perturbed descent scheme of the form

$$r_{t+1} = r_t + \gamma_t\big(s_t + w_t\big), \tag{3.3}$$

where $\gamma_t > 0$ is a step size, s_t is a search direction, and w_t is a perturbation term.

The directions s_t are assumed to satisfy the uniform descent conditions

$$C_1\|\nabla f(r_t)\|^2 \leq -\nabla f(r_t)^\top s_t, \tag{3.4}$$

$$\|s_t\| \leq C_2\|\nabla f(r_t)\|, \tag{3.5}$$

for some constants $C_1, C_2 > 0$. These conditions ensure that, in the absence of noise, the method possesses a uniform descent mechanism comparable to that of gradient-type algorithms.

We further assume that the perturbations satisfy the step-size–proportional bound

$$\|w_t\| \leq \gamma_t\big(q + p\|\nabla f(r_t)\|\big), \qquad \forall t, \tag{3.6}$$

for some constants $p, q \geq 0$. Finally, the step sizes are required to satisfy

$$\sum_{t=0}^{\infty} \gamma_t = \infty, \qquad \sum_{t=0}^{\infty} \gamma_t^2 < \infty. \tag{3.7}$$

The first condition in (3.7) ensures that the method does not stagnate prematurely, while the second guarantees that the cumulative effect of the vanishing perturbations remains finite.

Proposition 3.1 (Convergence under vanishing perturbations)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable, and suppose that ∇f is Lipschitz continuous. Assume that the iteration* (3.3) *satisfies* (3.4)– (3.7)*. Then one of the following alternatives holds:*

$$f(r_t) \to -\infty,$$

or $\{f(r_t)\}$ converges to a finite value and

$$\lim_{t\to\infty} \|\nabla f(r_t)\| = 0.$$

In the latter case, every limit point of the sequence $\{r_t\}$ is a stationary point of f.

Remark 3.2. The result shows that step-size–proportional perturbations are asymptotically negligible under the classical stochastic approximation conditions (3.7). The square-summability condition $\sum_t \gamma_t^2 < \infty$ ensures that the accumulated noise remains controlled, while the divergence of $\sum_t \gamma_t$ forces the iterates to continue making progress. These conditions are standard in the analysis of Robbins–Monro procedures and stochastic gradient methods. The proof, which relies on refined descent estimates and summability arguments, will be presented in a later section.

3.3 Stochastic Gradient Noise and Robbins–Monro Approximation

A fundamentally important class of perturbations arises from *random sampling*. In this setting, the noise is neither deterministic nor adversarial, but is generated by random variables with zero mean and finite variance. This framework underlies stochastic gradient methods and classical stochastic approximation theory.

3.3.1 Stochastic Gradient Model

Consider an objective function defined as an expectation:

$$f(r) = \mathbb{E}_v\big[F(r, v)\big], \tag{3.8}$$

where $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is continuously differentiable with respect to r, and v is a random vector defined on a probability space $(\Omega, \mathcal{A}, P)$.

Under standard regularity conditions ensuring interchange of differentiation and expectation (e.g., dominated convergence), the gradient of f is given by

$$\nabla f(r) = \mathbb{E}_v\big[\nabla_r F(r, v)\big]. \tag{3.9}$$

The exact gradient descent iteration therefore reads

$$r_{t+1} = r_t - \gamma_t \nabla f(r_t) = r_t - \gamma_t \mathbb{E}_v\big[\nabla_r F(r_t, v)\big].$$

In many applications, the expectation in (3.9) is computationally intractable. Instead, one draws an independent sample $v_t \sim P$ and defines the stochastic search direction

$$s_t = -\nabla_r F(r_t, v_t).$$

This leads to the stochastic gradient iteration

$$r_{t+1} = r_t - \gamma_t \nabla_r F(r_t, v_t). \tag{3.10}$$

3.3.2 Noise Decomposition

The iteration (3.10) can be written as a perturbed gradient method. Indeed,

$$\begin{aligned} r_{t+1} &= r_t - \gamma_t \left(\mathbb{E}_v\big[\nabla_r F(r_t, v)\big] + \underbrace{\nabla_r F(r_t, v_t) - \mathbb{E}_v\big[\nabla_r F(r_t, v)\big]}_{\triangleq w_t} \right) \\ &= r_t - \gamma_t \nabla f(r_t) + \gamma_t w_t. \end{aligned} \tag{3.11}$$

The random vector

$$w_t = \nabla_r F(r_t, v_t) - \mathbb{E}_v\big[\nabla_r F(r_t, v)\big] \tag{3.12}$$

satisfies

$$\mathbb{E}(w_t \mid \mathcal{F}_t) = 0,$$

where $\mathcal{F}_t$ denotes the sigma-algebra generated by $\{r_0, v_0, \ldots, v_{t-1}\}$. Thus, $\{w_t\}$ forms a *martingale difference sequence*. Moreover, if $\mathbb{E}(\|w_t\|^2 \mid \mathcal{F}_t) \leq \sigma^2$, the noise has uniformly bounded conditional variance.

Although the direction $-\nabla_r F(r_t, v_t)$ is not necessarily a descent direction for the deterministic objective f, it satisfies

$$\mathbb{E}\big[\nabla f(r_t)^\top s_t \mid \mathcal{F}_t\big] = -\|\nabla f(r_t)\|^2,$$

so the descent property holds *in conditional expectation*. This is the key structural feature enabling convergence analysis.

3.3.3 Robbins–Monro Stochastic Approximation

The stochastic gradient method is a special case of the classical Robbins–Monro stochastic approximation scheme. Consider the problem of finding a root of

$$h(r) = \mathbb{E}_v[g(r, v)] = 0, \tag{3.13}$$

where $g : \mathbb{R}^d \times \mathbb{R}^m \to \mathbb{R}^d$.

If the expectation were computable, one could apply the deterministic iteration

$$r_{t+1} = r_t - \gamma_t h(r_t).$$

When the distribution of v is unknown or the expectation is expensive to evaluate, it is replaced by a single sample:

$$r_{t+1} = r_t - \gamma_t g(r_t, v_t).$$

Writing

$$g(r_t, v_t) = h(r_t) + \big(g(r_t, v_t) - h(r_t)\big),$$

we obtain

$$r_{t+1} = r_t - \gamma_t \big(h(r_t) + w_t\big), \tag{3.14}$$

where w_t is a martingale difference noise term.

Algorithms of the form (3.14) are known as *Robbins–Monro stochastic approximation algorithms*. Under appropriate conditions on the step sizes, typically

$$\sum_{t=0}^{\infty} \gamma_t = \infty, \qquad \sum_{t=0}^{\infty} \gamma_t^2 < \infty,$$

and suitable stability assumptions on h, the iterates converge almost surely to a solution of (3.13).

Stochastic gradient descent corresponds to the special case $h(r) = \nabla f(r)$, so that minimizing an expected objective is equivalent to solving the stochastic approximation problem $\nabla f(r) = 0$.

3.4 ODE Method for Stochastic Approximation

We begin by stating the central result underlying the ODE method for stochastic approximation. This theorem formalizes the idea that a stochastic recursion tracks the trajectories of an associated deterministic ordinary differential equation and inherits its asymptotic behavior.

Theorem 3.3 (ODE method for stochastic approximation)**.** *Consider the stochastic approximation recursion*

$$X_{n+1} = X_n + \varepsilon_n \big(h(X_n) + M_n\big), \qquad X_0 \text{ given}, \tag{3.15}$$

where $h : \mathbb{R}^d \to \mathbb{R}^d$ is continuous, $\{M_n\}$ is a martingale difference sequence with respect to the natural filtration, and $\{\varepsilon_n\}$ is a deterministic step-size sequence satisfying

$$\sum_{n=0}^{\infty} \varepsilon_n = \infty, \qquad \sum_{n=0}^{\infty} \varepsilon_n^2 < \infty.$$

Assume further that the iterates $\{X_n\}$ are almost surely bounded and that the ordinary differential equation

$$\dot{X}(t) = h(X(t)) \tag{3.16}$$

admits a continuously differentiable Lyapunov function V such that

$$\langle \nabla V(x), h(x) \rangle \leq 0,$$

with equality if and only if $h(x) = 0$. Then, almost surely, every limit point of $\{X_n\}$ belongs to the invariant set

$$\mathcal{E} := \{x \in \mathbb{R}^d : h(x) = 0\},$$

and the stochastic recursion converges to $\mathcal{E}$.

The theorem provides a unifying framework for analyzing a wide class of stochastic approximation algorithms, including stochastic gradient methods and Robbins–Monro schemes. Its significance lies in the fact that the stochastic recursion (3.22) may be understood as a noisy, time-discretized approximation of the deterministic dynamical system (3.16). The diminishing step sizes ensure that the discretization becomes increasingly accurate, while the square-summability condition guarantees that the cumulative effect of the stochastic noise remains finite almost surely.

To make this connection precise, one introduces the continuous-time scale $t_n := \sum_{k=0}^{n-1} \varepsilon_k$ and constructs a piecewise linear interpolation $\bar{X}(t)$ of the iterates by setting $\bar{X}(t_n) = X_n$ and interpolating linearly on each interval $[t_n, t_{n+1}]$. For each n, let $X^{(n)}(t)$ denote the solution of the ODE (3.16) initialized at X_n at time t_n. Under the stated assumptions, one can show that for any fixed time horizon $T > 0$,

$$\sup_{t \in [t_n, t_n+T]} \|\bar{X}(t) - X^{(n)}(t)\| \longrightarrow 0 \quad \text{almost surely.}$$

This tracking property formalizes the intuition that, on finite time windows, the stochastic iterates evolve like trajectories of the limiting ODE, with the noise becoming negligible as the step size decreases.

Once this asymptotic tracking is established, the long-term behavior of the stochastic recursion is governed by the qualitative dynamics of the ODE (3.16). In particular, if the ODE admits a Lyapunov function with the stated properties, then all ODE trajectories converge to the invariant set $\mathcal{E}$. The stochastic recursion inherits this convergence almost surely, since it remains arbitrarily close to ODE trajectories over arbitrarily long finite horizons.

The ODE method thus reduces the analysis of stochastic approximation algorithms to the study of deterministic dynamical systems. Stability, attractivity, and invariance properties of the ODE directly determine the almost sure limit set of the stochastic algorithm. The martingale noise affects transient behavior but does not alter the asymptotic outcome, provided it is properly scaled by diminishing step sizes.

At a conceptual level, the proof of Theorem 3.23 proceeds by first establishing the tracking property using martingale convergence arguments and discrete-time Gronwall inequalities, and then invoking Lyapunov stability theory or invariance principles to characterize the limit set of the ODE. This two-stage structure—tracking followed by stability—lies at the heart of

modern stochastic approximation theory.

3.5 Mathematical Background and Preliminaries

We collect several probabilistic and analytical tools that will be used repeatedly in the convergence analysis of stochastic approximation algorithms.

3.5.1 Martingales

Definition 3.4 (Martingale)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{\mathcal{F}_n\}_{n\geq 0}$ a filtration. A sequence of random vectors $\{Y_n\}_{n\geq 0}$ is called a *martingale* with respect to $\{\mathcal{F}_n\}$ if:

1. Y_n is $\mathcal{F}_n$-measurable for all n,
2. $\mathbb{E}\big[\|Y_n\|\big] < \infty$ for all n,
3. $\mathbb{E}(Y_{n+1} \mid \mathcal{F}_n) = Y_n$ almost surely.

Example 3.5 (Partial sums of martingale differences)**.** Let $\{M_{n+1}\}$ be a sequence of random vectors such that

$$\mathbb{E}(M_{n+1} \mid \mathcal{F}_n) = 0, \qquad \mathbb{E}\big(\|M_{n+1}\|^2 \mid \mathcal{F}_n\big) < \infty.$$

Define

$$Y_n := \sum_{k=0}^{n-1} M_{k+1}.$$

Then $\{Y_n\}$ is a martingale with respect to $\{\mathcal{F}_n\}$.

This structure appears directly in stochastic approximation algorithms of the form

$$x_{n+1} = x_n + \varepsilon_n\big(h(x_n) + M_{n+1}\big),$$

where the cumulative noise term $\sum_{k=0}^{n-1} \varepsilon_k M_{k+1}$ is a scaled martingale.

Example 3.6 (Sample noise in stochastic gradient descent)**.** Let

$$f(x) = \mathbb{E}_v[F(x,v)], \qquad \nabla f(x) = \mathbb{E}_v[\nabla_x F(x,v)].$$

Suppose v_n are i.i.d. samples and define

$$M_{n+1} = \nabla_x F(x_n, v_n) - \nabla f(x_n).$$

Then

$$\mathbb{E}(M_{n+1} \mid \mathcal{F}_n) = 0,$$

so $\{M_{n+1}\}$ forms a martingale difference sequence. This is the canonical noise model underlying Robbins–Monro and stochastic gradient methods.

3.5.2 Martingale Convergence

Theorem 3.7 (Martingale Convergence Theorem)**.** *Let $\{Y_n\}$ be a martingale with respect to $\{\mathcal{F}_n\}$. Suppose that*

$$\sum_{n=0}^{\infty} \mathbb{E}\Big(\|Y_{n+1} - Y_n\|^2 \;\Big|\; \mathcal{F}_n\Big) < \infty \quad \textit{almost surely.} \tag{3.17}$$

Then there exists a finite random vector Y_∞ such that

$$Y_n \longrightarrow Y_\infty \quad \textit{almost surely.}$$

Proof. Define the martingale difference sequence

$$D_{n+1} := Y_{n+1} - Y_n, \qquad \mathbb{E}(D_{n+1} \mid \mathcal{F}_n) = 0.$$

Then

$$Y_n = Y_0 + \sum_{k=0}^{n-1} D_{k+1}.$$

Condition (3.17) implies that

$$\sum_{n=0}^{\infty} \mathbb{E}\Big(\|D_{n+1}\|^2\Big) < \infty.$$

By Kolmogorov's convergence theorem for vector-valued martingale difference sequences, the series $\sum_n D_{n+1}$ converges almost surely. Hence Y_n converges almost surely to a finite random vector Y_∞. □

Example 3.8 (Almost sure convergence of accumulated noise)**.** Let

$$S_n := \sum_{k=0}^{n-1} \varepsilon_k M_{k+1},$$

where $\{M_{n+1}\}$ is a martingale difference sequence and $\sum_{k=0}^{\infty} \varepsilon_k^2 < \infty$. Then

$$\sum_{k=0}^{\infty} \mathbb{E}\Big(\|\varepsilon_k M_{k+1}\|^2 \mid \mathcal{F}_k \Big) < \infty \quad \text{a.s.}$$

By Theorem 3.7, S_n converges almost surely.

This fact is essential in showing that stochastic approximation iterates track the limiting ODE asymptotically.

3.5.3 Gronwall Inequalities

Gronwall inequalities are fundamental tools for bounding solutions of differential and difference inequalities.

Theorem 3.9 (Gronwall Inequality: Differential Form)**.** *Let $x : [0, T] \to \mathbb{R}$ be a nonnegative absolutely continuous function satisfying*

$$\dot{x}(t) \leq L x(t), \qquad \forall t \in [0, T],$$

for some constant $L \geq 0$. Then

$$x(t) \leq x(0)\, e^{Lt}, \qquad \forall t \in [0, T].$$

Proof. Define $y(t) = e^{-Lt} x(t)$. Then

$$\dot{y}(t) = e^{-Lt}(\dot{x}(t) - L x(t)) \leq 0,$$

so $y(t) \leq y(0)$ for all t. Multiplying by e^{Lt} yields the result. □

Theorem 3.10 (Generalized Gronwall Inequality)**.** *Let $x : [0, T] \to \mathbb{R}$ be nonnegative and absolutely continuous, and suppose*

$$\dot{x}(t) \leq L(t) x(t),$$

where $L(\cdot)$ is a nonnegative integrable function. Then

$$x(t) \leq x(0) \exp\left(\int_0^t L(\tau)\, d\tau \right), \qquad \forall t \in [0, T].$$

Proof. The proof follows by integrating the inequality $\frac{d}{dt} \log x(t) \leq L(t)$ almost everywhere. □

Theorem 3.11 (Gronwall Inequality: Integral Form)**.** *Let $x : [0, T] \to \mathbb{R}$ be bounded and nonnegative, and suppose*

$$x(t) \leq B + \int_0^t L(\tau)x(\tau)\, d\tau, \qquad \forall t \in [0, T],$$

for some constant $B \geq 0$ and nonnegative integrable function L. Then

$$x(t) \leq B \exp\left(\int_0^t L(\tau)\, d\tau\right), \qquad \forall t \in [0, T].$$

Proof. Define

$$u(t) := B + \int_0^t L(\tau)x(\tau)\, d\tau.$$

Then $x(t) \leq u(t)$ and

$$\dot{u}(t) = L(t)x(t) \leq L(t)u(t).$$

Applying Theorem 3.10 to u yields the desired bound. □

Example 3.12 (Stability of linear ODEs)**.** Consider the scalar ODE

$$\dot{x}(t) = a(t)x(t), \qquad a(t) \leq L.$$

Then

$$x(t) \leq x(0) \exp\left(\int_0^t a(\tau)\, d\tau\right) \leq x(0)e^{Lt}.$$

This is a direct application of the differential Gronwall inequality and is used to bound the growth of error trajectories in ODE approximations.

Example 3.13 (Perturbation of ODE solutions)**.** Let $x(t)$ and $\tilde{x}(t)$ satisfy

$$\dot{x}(t) = h(x(t)), \qquad \dot{\tilde{x}}(t) = h(\tilde{x}(t)) + \delta(t),$$

with $\|\delta(t)\| \leq \varepsilon$. Then

$$\|x(t) - \tilde{x}(t)\| \leq \varepsilon t\, e^{Lt},$$

where L is the Lipschitz constant of h. This estimate underlies the ODE method for stochastic approximation.

3.5.4 Discrete-Time Gronwall Inequality

We now state a discrete analogue that plays a central role in analyzing recursive algorithms.

Lemma 3.14 (Discrete-Time Gronwall Inequality)**.** *Let $K \geq 0$, and let $\{x_n\}_{n\geq 0}$ and $\{a_n\}_{n\geq 0}$ be nonnegative sequences satisfying*

$$x_{n+1} \leq K + \sum_{m=0}^{n} a_m x_m, \qquad \forall n \geq 0.$$

Then, for all $n \geq 0$,

$$x_n \leq K \exp\left(\sum_{m=0}^{n-1} a_m\right).$$

Proof. Define

$$S_n := \sum_{m=0}^{n-1} a_m, \qquad y_n := x_n e^{-S_n}.$$

Then

$$y_{n+1} = x_{n+1} e^{-S_{n+1}} \leq K e^{-S_{n+1}} + \sum_{m=0}^{n} a_m x_m e^{-S_{n+1}} = K e^{-S_{n+1}} + \sum_{m=0}^{n} a_m y_m e^{-(S_{n+1}-S_m)}.$$

Since $S_{n+1} - S_m \geq 0$, the sum is bounded by $\sum_{m=0}^{n} a_m y_m$. Iterating yields $y_n \leq K$, which implies the desired bound. □

Remark 3.15. Lemma 3.14 is the discrete counterpart of the integral Gronwall inequality and is a fundamental tool for controlling error growth in numerical schemes, ODE approximations, and stochastic approximation algorithms.

Example 3.16 (Bounding recursive error sequences)**.** Suppose a sequence $\{x_n\}$ satisfies

$$x_{n+1} \leq K + \sum_{m=0}^{n} a_m x_m, \qquad a_m \geq 0.$$

Such inequalities arise when comparing a numerical iterate to an exact ODE solution.

Lemma 3.14 implies

$$x_n \leq K \exp\left(\sum_{m=0}^{n-1} a_m\right),$$

providing a uniform bound on error growth.

Example 3.17 (Tracking error in stochastic approximation)**.** In the ODE method, one encounters bounds of the form

$$\|x^{(n)}(t_k) - x_k\| \leq \sum_{\ell=n}^{k-1} \varepsilon_\ell L \|x^{(n)}(t_\ell) - x_\ell\| + \left\| \sum_{\ell=n}^{k-1} \varepsilon_\ell M_{\ell+1} \right\|.$$

Applying Lemma 3.14 allows one to control the tracking error uniformly over finite time win-

dows.

3.6 Stochastic Approximation Framework

We now present a standard convergence framework for stochastic approximation algorithms based on the ODE method.

Consider the stochastic recursion

$$x_{n+1} = x_n + \varepsilon_n \big(h(x_n) + M_{n+1}\big), \qquad x_0 \text{ given}, \tag{3.18}$$

where $\{x_n\} \subset \mathbb{R}^d$, $h : \mathbb{R}^d \to \mathbb{R}^d$, $\{\varepsilon_n\}$ is a step-size sequence, and $\{M_n\}$ is a noise process adapted to a filtration $\{\mathcal{F}_n\}$.

Assumption 3.18 (A1: Lipschitz continuity of h)**.** There exists $L > 0$ such that

$$\|h(x) - h(y)\| \le L\|x - y\|, \qquad \forall x, y \in \mathbb{R}^d.$$

Assumption 3.19 (A2: Diminishing step sizes)**.** The step sizes satisfy

$$\sum_{n=0}^{\infty} \varepsilon_n = \infty, \qquad \sum_{n=0}^{\infty} \varepsilon_n^2 < \infty, \qquad \varepsilon_n \to 0.$$

Assumption 3.20 (A3: Martingale difference noise)**.** There exists $K > 0$ such that, for all n,

$$\mathbb{E}(M_{n+1} \mid \mathcal{F}_n) = 0, \qquad \mathbb{E}(\|M_{n+1}\|^2 \mid \mathcal{F}_n) \le K\big(1 + \|x_n\|^2\big),$$

where

$$\mathcal{F}_n = \sigma(x_0, M_1, \ldots, x_n, M_n).$$

Assumption 3.21 (A4: Bounded iterates)**.**

$$\sup_{n \ge 0} \|x_n\| < \infty \quad \text{almost surely.}$$

Assumption 3.22 (A5: Lyapunov stability)**.** There exists a continuously differentiable, positive, radially unbounded function $V : \mathbb{R}^d \to \mathbb{R}$ such that

$$\langle \nabla V(x), h(x) \rangle \le 0, \qquad \forall x \in \mathbb{R}^d,$$

with strict inequality whenever $x \notin \mathcal{S}, \quad \mathcal{S} := \{x : \nabla V(x) = 0\}$.

Theorem 3.23 (ODE Method for Stochastic Approximation)**.** *Under assumptions* (A1)–(A5)*, the sequence* $\{x_n\}$ *generated by* (3.18) *satisfies*

$$V(x_n) \longrightarrow V_\infty \quad \textit{almost surely},$$

for some random limit V_∞*, and every limit point of* $\{x_n\}$ *belongs almost surely to the internally chain transitive set of the ODE*

$$\dot{x} = h(x).$$

If, in addition, the ODE has a globally asymptotically stable equilibrium at the origin, then

$$x_n \longrightarrow 0 \quad \textit{almost surely}.$$

Proof. The proof proceeds in three steps.

Step 1: Continuous-time interpolation. Define the continuous-time scale

$$t_n := \sum_{k=0}^{n-1} \varepsilon_k, \qquad t_0 = 0.$$

By Assumption (A2), $t_n \to \infty$.

Define the piecewise linear interpolation $\bar{x}(t)$ of the iterates by

$$\bar{x}(t_n) = x_n, \qquad \bar{x}(t) = x_n + \frac{t - t_n}{\varepsilon_n}(x_{n+1} - x_n), \quad t \in [t_n, t_{n+1}].$$

For each n, let $x^{(n)}(t)$ denote the solution of the ODE

$$\dot{x}(t) = h(x(t)),$$

with initial condition

$$x^{(n)}(t_n) = x_n.$$

The key step is to show that $\bar{x}(t)$ asymptotically tracks the ODE trajectory $x^{(n)}(t)$ on finite time horizons.

Step 2: Proof of the tracking property. Fix $T > 0$ and define

$$m := \inf\{k \geq n : t_k \geq t_n + T\}.$$

For any $n \leq k \leq m$, the ODE solution satisfies

$$\begin{aligned} x^{(n)}(t_k) &= x_n + \int_{t_n}^{t_k} h(x^{(n)}(s))\, ds \\ &= x_n + \sum_{\ell=n}^{k-1} \int_{t_\ell}^{t_{\ell+1}} h(x^{(n)}(s))\, ds. \end{aligned}$$

The stochastic recursion yields

$$x_k = x_n + \sum_{\ell=n}^{k-1} \varepsilon_\ell h(x_\ell) + \sum_{\ell=n}^{k-1} \varepsilon_\ell M_{\ell+1}.$$

Subtracting the two expressions gives

$$\begin{aligned} \|x^{(n)}(t_k) - x_k\| \leq \sum_{\ell=n}^{k-1} \varepsilon_\ell \|h(x^{(n)}(t_\ell)) - h(x_\ell)\| & \quad (3.19) \\ + \left\| \sum_{\ell=n}^{k-1} \varepsilon_\ell M_{\ell+1} \right\| & \quad (3.20) \\ + \sum_{\ell=n}^{k-1} \int_{t_\ell}^{t_{\ell+1}} \|h(x^{(n)}(s)) - h(x^{(n)}(t_\ell))\|\, ds. & \quad (3.21) \end{aligned}$$

Term (3.19)**.** By Assumption (A1),

$$\|h(x^{(n)}(t_\ell)) - h(x_\ell)\| \leq L\|x^{(n)}(t_\ell) - x_\ell\|.$$

Term (3.20)**.** Define

$$S_k := \sum_{\ell=0}^{k-1} \varepsilon_\ell M_{\ell+1}.$$

By Assumptions (A2)–(A4),

$$\sum_{n=0}^{\infty} \mathbb{E}\big(\|S_{n+1} - S_n\|^2 \mid \mathcal{F}_n\big) = \sum_{n=0}^{\infty} \varepsilon_n^2 \mathbb{E}(\|M_{n+1}\|^2 \mid \mathcal{F}_n) < \infty \quad \text{a.s.}$$

Hence, by the martingale convergence theorem,

$$S_k \to S_\infty \quad \text{almost surely.}$$

Therefore,

$$\sup_{n\le k\le m}\left\|\sum_{\ell=n}^{k-1}\varepsilon_\ell M_{\ell+1}\right\| \longrightarrow 0 \quad \text{a.s.}$$

Term (3.21)**.** Using Lipschitz continuity of h,

$$\|h(x^{(n)}(s)) - h(x^{(n)}(t_\ell))\| \le L\|x^{(n)}(s) - x^{(n)}(t_\ell)\|.$$

Since ODE solutions are locally Lipschitz on compact intervals and $\varepsilon_\ell \to 0$,

$$\sum_{\ell=n}^{k-1}\int_{t_\ell}^{t_{\ell+1}} \|h(x^{(n)}(s)) - h(x^{(n)}(t_\ell))\|\, ds = \mathcal{O}\left(\sum_{\ell=n}^{k-1}\varepsilon_\ell^2\right) \longrightarrow 0.$$

Conclusion of Step 2. Combining all terms, we obtain

$$\|x^{(n)}(t_k) - x_k\| \le \sum_{\ell=n}^{k-1}\varepsilon_\ell L\|x^{(n)}(t_\ell) - x_\ell\| + o(1).$$

Applying a discrete Gronwall inequality yields

$$\sup_{n\le k\le m}\|x^{(n)}(t_k) - x_k\| \longrightarrow 0 \quad \text{a.s.}$$

By linear interpolation,

$$\sup_{t\in[t_n,t_n+T]}\|x^{(n)}(t) - \bar{x}(t)\| \longrightarrow 0 \quad \text{a.s.}$$

which establishes the tracking property.

Step 3: Lyapunov convergence. By Assumption (A5),

$$\frac{d}{dt}V(x(t)) = \langle \nabla V(x(t)), h(x(t))\rangle \le 0,$$

with strict inequality outside the invariant set

$$\mathcal{S} := \{x : \nabla V(x) = 0\}.$$

Thus, all ODE trajectories approach $\mathcal{S}$.

By the tracking property, $\bar{x}(t)$ remains asymptotically close to ODE trajectories on every finite horizon. Hence,

$$V(x_n) = V(\bar{x}(t_n)) \longrightarrow V_\infty \quad \text{a.s.}$$

and all limit points of $\{x_n\}$ belong to $\mathcal{S}$.

Therefore, x_n converges almost surely to the invariant set of the ODE $\dot{x} = h(x)$, completing the proof. □

3.7 Results for Stochastic Gradient and Robbins–Monro Algorithms

The stochastic gradient iteration (**??**) and the fixed-point recursion (**??**) both fall within the general framework of stochastic approximation:

$$r_{t+1} = r_t + \gamma_t \big(h(r_t) + w_t\big), \tag{3.22}$$

where:

$$h(r) = \begin{cases} -\nabla f(r), & \text{(stochastic gradient)} \\ Hr - r, & \text{(fixed-point / Robbins–Monro)} \end{cases}$$

and $\{w_t\}$ is a martingale difference noise sequence adapted to a filtration $\{\mathcal{F}_t\}$.

This formulation allows us to apply the ODE method and martingale convergence results developed earlier.

3.7.1 Standing Assumptions

Assume the following:

(SG1) (*Smoothness*) f is continuously differentiable and ∇f is Lipschitz:

$$\|\nabla f(r) - \nabla f(\bar{r})\| \leq L\|r - \bar{r}\|.$$

(SG2) (*Step sizes*)

$$\sum_{t=0}^{\infty} \gamma_t = \infty, \qquad \sum_{t=0}^{\infty} \gamma_t^2 < \infty, \qquad \gamma_t \to 0.$$

(SG3) (*Martingale noise*)

$$\mathbb{E}(w_t \mid \mathcal{F}_t) = 0, \qquad \mathbb{E}(\|w_t\|^2 \mid \mathcal{F}_t) \leq C(1 + \|r_t\|^2).$$

(SG4) (*Stability*)

$$\sup_t \|r_t\| < \infty \quad \text{almost surely.}$$

These assumptions are standard in stochastic approximation and are satisfied by a broad class of stochastic gradient and learning algorithms.

3.7.2 ODE Interpretation

Under the above assumptions, the stochastic recursion (3.22) asymptotically tracks the solution of the limiting ODE

$$\dot{r}(t) = h(r(t)). \tag{3.23}$$

In particular:

- For stochastic gradient descent,

$$\dot{r}(t) = -\nabla f(r(t)).$$

- For Robbins–Monro fixed-point iterations,

$$\dot{r}(t) = Hr(t) - r(t).$$

3.7.3 Main Convergence Result

Theorem 3.24 (Almost sure convergence of stochastic gradient methods)**.** *Under assumptions* (SG1)–(SG4)*, the iterates generated by* (3.10) *satisfy:*

1. *Either $f(r_t) \to -\infty$, or $\{f(r_t)\}$ converges to a finite value almost surely.*
2. $$\lim_{t\to\infty} \nabla f(r_t) = 0 \quad \textit{almost surely}.$$
3. *Every limit point of $\{r_t\}$ is a stationary point of f.*

Proof. The stochastic gradient recursion can be written as

$$r_{t+1} = r_t - \gamma_t \nabla f(r_t) + \gamma_t w_t, \tag{3.24}$$

where $\{w_t\}$ is a martingale difference sequence satisfying

$$\mathbb{E}(w_t \mid \mathcal{F}_t) = 0, \qquad \mathbb{E}(\|w_t\|^2 \mid \mathcal{F}_t) \le C(1 + \|r_t\|^2).$$

Step 1: Descent inequality. Since ∇f is Lipschitz continuous with constant L, the quadratic upper bound yields

$$f(r_{t+1}) \leq f(r_t) + \nabla f(r_t)^\top (r_{t+1} - r_t) + \frac{L}{2}\|r_{t+1} - r_t\|^2.$$

Substituting (3.24) gives

$$\begin{aligned} f(r_{t+1}) \leq f(r_t) - \gamma_t \|\nabla f(r_t)\|^2 + \gamma_t \nabla f(r_t)^\top w_t \\ + \frac{L}{2}\gamma_t^2 \|\nabla f(r_t) + w_t\|^2. \end{aligned}$$

Step 2: Conditional expectation. Taking conditional expectation with respect to $\mathcal{F}_t$ and using $\mathbb{E}(w_t \mid \mathcal{F}_t) = 0$,

$$\mathbb{E}\big[f(r_{t+1}) \mid \mathcal{F}_t\big] \leq f(r_t) - \gamma_t \|\nabla f(r_t)\|^2 + C'\gamma_t^2 (1 + \|\nabla f(r_t)\|^2),$$

for some constant $C' > 0$, where boundedness of $\{r_t\}$ has been used.

Step 3: Almost sure summability. Summing over t and using $\sum_t \gamma_t^2 < \infty$, the Robbins–Siegmund supermartingale convergence theorem implies

$$\sum_{t=0}^{\infty} \gamma_t \|\nabla f(r_t)\|^2 < \infty \quad \text{a.s.}$$

Since $\sum_t \gamma_t = \infty$, this implies

$$\liminf_{t\to\infty} \|\nabla f(r_t)\| = 0 \quad \text{a.s.}$$

Step 4: Full convergence of the gradient. Suppose, by contradiction, that $\limsup_{t\to\infty} \|\nabla f(r_t)\| > 0$. Then there exists $\varepsilon > 0$ and a subsequence $\{t_k\}$ such that $\|\nabla f(r_{t_k})\| \geq \varepsilon$ for all k. Since $\gamma_t > 0$ and $\sum_t \gamma_t = \infty$, the summability $\sum_t \gamma_t \|\nabla f(r_t)\|^2 < \infty$ would be violated. Hence,

$$\lim_{t\to\infty} \nabla f(r_t) = 0 \quad \text{a.s.}$$

Step 5: Limit points. Let $\bar{r}$ be a limit point of $\{r_t\}$. By continuity of ∇f,

$$\nabla f(\bar{r}) = 0,$$

so every limit point is stationary.

Finally, since $f(r_{t+1}) \leq f(r_t) + \mathcal{O}(\gamma_t^2)$, and $\sum_t \gamma_t^2 < \infty$, either $f(r_t) \to -\infty$ or $\{f(r_t)\}$ converges to a finite value. □

3.7.4 Consequences for Convex Objectives

Corollary 3.25 (Convergence for convex stochastic optimization)**.** *If, in addition, f is convex, then the iterates r_t converge almost surely to the set of global minimizers of f. If f is strongly convex, the convergence is to the unique minimizer $r^\star$.*

Proof. If f is convex, every stationary point is a global minimizer. Since all limit points of $\{r_t\}$ are stationary by Theorem 3.24, the result follows.

If f is strongly convex, the minimizer is unique. Hence all convergent subsequences of $\{r_t\}$ share the same limit, implying $r_t \to r^\star$ almost surely. □

3.7.5 Convergence of Robbins–Monro Algorithms

Theorem 3.26 (Robbins–Monro convergence)**.** *Consider the stochastic approximation scheme*

$$r_{t+1} = (1 - \gamma_t) r_t + \gamma_t (H r_t + w_t),$$

where H is a contraction mapping with modulus $\rho < 1$ and $\{w_t\}$ is martingale difference noise. Under assumptions (SG2)–(SG4)*,*

$$r_t \to r^\star \quad \textit{almost surely},$$

where $r^\star$ is the unique solution of the fixed-point equation $Hr = r$.

Proof. Rewrite the recursion as

$$r_{t+1} - r^\star = (1 - \gamma_t)(r_t - r^\star) + \gamma_t (H r_t - H r^\star) + \gamma_t w_t.$$

Using contraction of H,

$$\|H r_t - H r^\star\| \leq \rho \|r_t - r^\star\|.$$

Hence,

$$\|r_{t+1} - r^\star\| \leq \big(1 - \gamma_t (1 - \rho)\big) \|r_t - r^\star\| + \gamma_t \|w_t\|.$$

Taking conditional expectations and applying the Robbins–Siegmund theorem, together

with $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 < \infty$, yields

$$r_t \to r^\star \quad \text{almost surely.}$$

□

Remark 3.27. These results show that stochastic gradient and Robbins–Monro algorithms asymptotically behave like their deterministic counterparts:

- stochastic gradient descent tracks the gradient flow $\dot{r} = -\nabla f(r)$,
- Robbins–Monro iterations track stable fixed-point dynamics.

Martingale noise is averaged out provided step sizes are diminishing and square-summable.

3.8 Illustrative Examples

We now present concrete examples illustrating how classical stochastic algorithms fit into the stochastic approximation framework (3.18) and satisfy Assumptions (A1)–(A5).

3.8.1 Example 1: Stochastic Gradient Descent for Strongly Convex Functions

Let $f : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable, m-strongly convex, and with Lipschitz gradient. Suppose that instead of the true gradient, we observe a noisy estimate

$$g(x, \xi),$$

where ξ is a random sample and

$$\mathbb{E}[g(x, \xi)] = \nabla f(x).$$

The stochastic gradient iteration is

$$x_{n+1} = x_n - \varepsilon_n g(x_n, \xi_{n+1}).$$

Writing

$$M_{n+1} = -\big(g(x_n, \xi_{n+1}) - \nabla f(x_n)\big),$$

the recursion becomes

$$x_{n+1} = x_n + \varepsilon_n\big(-\nabla f(x_n) + M_{n+1}\big),$$

which is of the form (3.18) with

$$h(x) = -\nabla f(x).$$

Verification of assumptions.

- (A1) holds because ∇f is Lipschitz.
- (A2) is satisfied for step sizes such as $\varepsilon_n = 1/(n+1)$.
- (A3) holds since M_{n+1} is a martingale difference with bounded conditional variance under standard moment assumptions.
- (A4) follows from strong convexity and coercivity of f.
- (A5) holds with Lyapunov function $V(x) = f(x) - f(x^\star)$, since

$$\langle \nabla V(x), h(x) \rangle = -\|\nabla f(x)\|^2 \leq 0.$$

The limiting ODE is

$$\dot{x} = -\nabla f(x),$$

which has a globally asymptotically stable equilibrium at $x^\star$. Hence $x_n \to x^\star$ almost surely.

3.8.2 Example 2: Robbins–Monro Root-Finding

Let $H : \mathbb{R}^d \to \mathbb{R}^d$ be a contraction mapping with fixed point $x^\star$. Suppose we observe noisy measurements

$$\tilde{H}(x, \xi) = H(x) + \text{noise}.$$

The Robbins–Monro iteration is

$$x_{n+1} = x_n + \varepsilon_n \big(\tilde{H}(x_n, \xi_{n+1}) - x_n\big).$$

Writing

$$h(x) = H(x) - x, \qquad M_{n+1} = \tilde{H}(x_n, \xi_{n+1}) - H(x_n),$$

we obtain the standard SA form.

The limiting ODE is

$$\dot{x} = H(x) - x.$$

Since H is a contraction,

$$\langle x - x^\star, h(x) \rangle \leq -(1-\rho)\|x - x^\star\|^2,$$

so the equilibrium $x^\star$ is globally asymptotically stable.

A Lyapunov function is

$$V(x) = \frac{1}{2}\|x - x^\star\|^2.$$

Thus the Robbins–Monro iteration converges almost surely to $x^\star$.

3.8.3 Example 3: Linear Stochastic Approximation

Consider

$$h(x) = -Ax,$$

where A is positive definite. The recursion becomes

$$x_{n+1} = x_n - \varepsilon_n A x_n + \varepsilon_n M_{n+1}.$$

The limiting ODE is

$$\dot{x} = -Ax,$$

whose solution is

$$x(t) = e^{-At}x(0),$$

and which converges exponentially to zero.

Take Lyapunov function

$$V(x) = \frac{1}{2}x^\top x.$$

Then

$$\langle \nabla V(x), h(x) \rangle = -x^\top A x \leq -\lambda_{\min}(A)\|x\|^2.$$

Hence the origin is globally asymptotically stable, and under the SA assumptions we obtain

$$x_n \to 0 \quad \text{almost surely.}$$

4 Newton's Method

Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable. Newton's method is a second–order optimization algorithm derived from a local quadratic approximation of the objective function.

4.1 Newton Direction and Quadratic Model

Given an iterate $x_k \in \mathbb{R}^n$, the *pure Newton iteration* is defined by

$$x_{k+1} = x_k - \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k), \tag{4.1}$$

provided the Hessian $\nabla^2 f(x_k)$ is invertible. The corresponding *Newton direction* is

$$d_k = -\left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k). \tag{4.2}$$

The Newton direction arises as the unique minimizer of the quadratic Taylor model of f at x_k,

$$m_k(d) := f(x_k) + \nabla f(x_k)^\top d + \frac{1}{2} d^\top \nabla^2 f(x_k) d. \tag{4.3}$$

Indeed, differentiating $m_k(d)$ with respect to d and setting the result equal to zero yields

$$\nabla m_k(d) = \nabla f(x_k) + \nabla^2 f(x_k) d = 0,$$

which gives $d = d_k$ whenever $\nabla^2 f(x_k)$ is nonsingular. If, in addition, $\nabla^2 f(x_k) \succ 0$, then d_k is the unique global minimizer of m_k.

Definition 4.1 (Newton decrement)**.** The *Newton decrement* at x_k is defined as

$$\lambda(x_k) := \left(\nabla f(x_k)^\top \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k)\right)^{1/2}. \tag{4.4}$$

Evaluating the quadratic model (4.3) at the minimizer d_k yields

$$m_k(0) - m_k(d_k) = -\nabla f(x_k)^\top d_k - \frac{1}{2} d_k^\top \nabla^2 f(x_k) d_k.$$

Using $\nabla^2 f(x_k) d_k = -\nabla f(x_k)$, we obtain

$$m_k(0) - m_k(d_k) = \frac{1}{2} \nabla f(x_k)^\top \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k) = \frac{1}{2} \lambda(x_k)^2.$$

Thus $\lambda(x_k)^2/2$ represents the predicted decrease in the quadratic model.

Moreover,

$$\lambda(x_k)^2 = d_k^\top \nabla^2 f(x_k) d_k = \|d_k\|^2_{\nabla^2 f(x_k)},$$

so the Newton decrement is precisely the norm of the Newton step in the local Hessian metric. When $\nabla^2 f(x_k) \succ 0$, the condition $\lambda(x_k) = 0$ is equivalent to $\nabla f(x_k) = 0$.

4.2 Newton's Method with Backtracking Line Search

To ensure global convergence from arbitrary initial points, the pure Newton iteration is combined with a backtracking line search. This yields a damped Newton method that guarantees sufficient decrease at every iteration.

Let $\varepsilon > 0$, $\sigma \in (0, 1/2)$, and $\beta \in (0, 1)$. Given an initial point $x_0 \in \mathbb{R}^n$, generate the sequence $\{x_k\}$ as follows.

Algorithm: Newton's Method with Backtracking Line Search

Input: $x_0 \in \mathbb{R}^n$, parameters $\varepsilon > 0$, $\sigma \in (0, 1/2)$, $\beta \in (0, 1)$.

For $k = 0, 1, 2, \ldots$ *repeat:*

Step 1 (Newton direction). Compute

$$d_k = -\left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k),$$

assuming that $\nabla^2 f(x_k)$ is nonsingular.

Step 2 (Newton decrement). Compute

$$\lambda(x_k)^2 = \nabla f(x_k)^\top \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k).$$

Step 3 (Termination test). If

$$\frac{\lambda(x_k)^2}{2} \leq \varepsilon,$$

terminate and return x_k. This condition corresponds to a sufficiently small predicted decrease in the quadratic model.

Step 4 (Backtracking line search). Initialize $\alpha_k = 1$. Reduce $\alpha_k \leftarrow \beta\alpha_k$ until the Armijo sufficient decrease condition

$$f(x_k + \alpha_k d_k) \leq f(x_k) + \sigma\alpha_k \nabla f(x_k)^\top d_k$$

is satisfied.

Step 5 (Update). Set

$$x_{k+1} = x_k + \alpha_k d_k.$$

End For

4.3 Global and Local Convergence Analysis

Newton's method exhibits two distinct regimes: a *damped global phase*, ensuring descent from arbitrary initial points, and a *local quadratic phase* once the iterates enter a neighborhood of the minimizer.

Let $L_0 := \{x \in \mathbb{R}^n \mid f(x) \leq f(x_0)\}$ denote the level set determined by the initial point x_0.

Assumption 4.2 (Regularity and curvature)**.** The following properties hold on L_0:

1. The level set L_0 is closed and bounded.
2. f is strongly convex with parameter $m > 0$:

$$\nabla^2 f(x) \succeq mI, \qquad \forall x \in L_0.$$

3. The Hessian is Lipschitz continuous with constant $L > 0$:

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq L\|x - y\|, \qquad \forall x, y \in L_0.$$

4. The Hessian is uniformly bounded above:

$$\nabla^2 f(x) \preceq MI, \qquad \forall x \in L_0.$$

Under Assumption 4.2, the function f admits a unique minimizer $x^\star \in L_0$.

Theorem 4.3 (Global convergence: damped Newton phase)**.** *Under Assumption 4.2, there exists a constant*

$$\eta \in \left(0, \frac{2m^2}{L}\right)$$

such that if

$$\|\nabla f(x_k)\| \geq \eta,$$

then the backtracking line search accepts a step size satisfying

$$\alpha_k \geq \beta \frac{m}{M},$$

and the objective decrease satisfies

$$f(x_{k+1}) - f(x_k) \leq -\gamma, \qquad \gamma := \sigma\beta \frac{m}{M^2}\eta^2. \tag{4.5}$$

Consequently, this phase terminates after finitely many iterations.

Proof. Let

$$d_k = -[\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$$

be the Newton direction.

Since $\nabla^2 f(x) \preceq MI$ on L_0, the standard quadratic upper bound gives, for all $\alpha \in [0, 1]$,

$$f(x_k + \alpha d_k) \leq f(x_k) + \alpha \nabla f(x_k)^\top d_k + \frac{\alpha^2 M}{2} \|d_k\|^2. \tag{4.6}$$

First observe

$$\nabla f(x_k)^\top d_k = -\nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k) = -\lambda(x_k)^2.$$

Strong convexity implies

$$\nabla^2 f(x_k) \succeq mI \quad \Rightarrow \quad \|d_k\| \leq \frac{1}{m} \|\nabla f(x_k)\|,$$

and also

$$\|d_k\|^2 \leq \frac{1}{m} \lambda(x_k)^2.$$

Substituting into (4.6) yields

$$f(x_k + \alpha d_k) \leq f(x_k) - \alpha \lambda(x_k)^2 + \frac{\alpha^2 M}{2m} \lambda(x_k)^2 = f(x_k) - \alpha \left(1 - \frac{\alpha M}{2m}\right) \lambda(x_k)^2.$$

Choosing

$$\alpha = \frac{m}{M}$$

gives

$$f(x_k + \alpha d_k) \leq f(x_k) - \frac{m}{2M}\lambda(x_k)^2.$$

Since $\sigma < \frac{1}{2}$, this implies the Armijo condition

$$f(x_k + \alpha d_k) \leq f(x_k) + \sigma\alpha\nabla f(x_k)^\top d_k.$$

Hence the backtracking line search accepts some

$$\alpha_k \geq \beta\frac{m}{M}.$$

Using

$$\lambda(x_k)^2 = \nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1}\nabla f(x_k) \geq \frac{1}{M}\|\nabla f(x_k)\|^2,$$

we obtain

$$f(x_{k+1}) - f(x_k) \leq -\sigma\alpha_k\lambda(x_k)^2 \leq -\sigma\beta\frac{m}{M^2}\|\nabla f(x_k)\|^2.$$

If $\|\nabla f(x_k)\| \geq \eta$, then (4.5) follows immediately.

Since f is bounded below on L_0, only finitely many decreases of magnitude at least γ can occur. Thus the damped phase terminates in finite time. □

Theorem 4.4 (Local quadratic convergence)**.** *Under Assumption 4.2, if*

$$\|\nabla f(x_k)\| < \eta,$$

then the full Newton step ($\alpha_k = 1$) is accepted and

$$\|\nabla f(x_{k+1})\| \leq \frac{L}{2m^2}\|\nabla f(x_k)\|^2. \tag{4.7}$$

Proof. The proof proceeds as follows.

Step 1: Second-order expansion of the gradient. By the mean-value form of Taylor's theorem,

$$\nabla f(x_k + d_k) = \nabla f(x_k) + \int_0^1 \nabla^2 f(x_k + td_k)\, d_k\, dt.$$

Add and subtract $\nabla^2 f(x_k)d_k$ inside the integral to obtain

$$\begin{aligned}\nabla f(x_k + d_k) &= \nabla f(x_k) + \nabla^2 f(x_k)d_k \\ &\quad + \int_0^1 \big(\nabla^2 f(x_k + td_k) - \nabla^2 f(x_k)\big)d_k\, dt.\end{aligned}$$

Since d_k is the Newton direction,

$$\nabla f(x_k) + \nabla^2 f(x_k)d_k = 0,$$

and therefore

$$\nabla f(x_k + d_k) = \int_0^1 \big(\nabla^2 f(x_k + td_k) - \nabla^2 f(x_k)\big)d_k\, dt.$$

Step 2: Use of Lipschitz continuity of the Hessian. By Assumption 4.2,

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \le L\|x - y\|.$$

Hence, for each $t \in [0, 1]$,

$$\|\nabla^2 f(x_k + td_k) - \nabla^2 f(x_k)\| \le Lt\|d_k\|.$$

Therefore,

$$\begin{aligned}\|\nabla f(x_k + d_k)\| &\le \int_0^1 \|\nabla^2 f(x_k + td_k) - \nabla^2 f(x_k)\|\, \|d_k\|\, dt \\ &\le \int_0^1 Lt\|d_k\|^2\, dt = \frac{L}{2}\|d_k\|^2.\end{aligned}$$

This yields

$$\|\nabla f(x_{k+1})\| \le \frac{L}{2}\|d_k\|^2.$$

Step 3: Bounding the Newton step. Strong convexity implies

$$\nabla^2 f(x_k) \succeq mI,$$

so

$$\|[\nabla^2 f(x_k)]^{-1}\| \le \frac{1}{m}.$$

Hence,

$$\|d_k\| = \|[\nabla^2 f(x_k)]^{-1}\nabla f(x_k)\| \leq \frac{1}{m}\|\nabla f(x_k)\|.$$

Substituting into the previous inequality gives

$$\|\nabla f(x_{k+1})\| \leq \frac{L}{2m^2}\|\nabla f(x_k)\|^2,$$

which establishes (4.7).

Step 4: Acceptance of the full Newton step. It remains to show that the full step $\alpha_k = 1$ satisfies the Armijo condition.

From the quadratic upper bound (cf. (4.6) with $\alpha = 1$),

$$f(x_k + d_k) \leq f(x_k) + \nabla f(x_k)^\top d_k + \frac{M}{2}\|d_k\|^2.$$

Since

$$\nabla f(x_k)^\top d_k = -\lambda(x_k)^2,$$

and

$$\|d_k\|^2 \leq \frac{1}{m^2}\|\nabla f(x_k)\|^2 \leq \frac{1}{m}\lambda(x_k)^2,$$

we obtain

$$f(x_k + d_k) \leq f(x_k) - \lambda(x_k)^2 + \frac{M}{2m}\lambda(x_k)^2 = f(x_k) - \left(1 - \frac{M}{2m}\right)\lambda(x_k)^2.$$

If $\|\nabla f(x_k)\| < \eta$ with

$$\eta < \frac{2m^2}{L},$$

then the quadratic estimate

$$\|\nabla f(x_{k+1})\| \leq \frac{L}{2m^2}\|\nabla f(x_k)\|^2 < \|\nabla f(x_k)\|$$

shows that the Newton step produces sufficient decrease.

More precisely, since

$$\nabla f(x_k)^\top d_k = -\lambda(x_k)^2 < 0,$$

and the higher-order term is quadratic in $\|d_k\|$, for sufficiently small $\|\nabla f(x_k)\|$ the inequality

$$f(x_k + d_k) \leq f(x_k) + \sigma \nabla f(x_k)^\top d_k$$

holds for any fixed $\sigma \in (0, \frac{1}{2})$. Thus the Armijo condition is satisfied with $\alpha_k = 1$.

□

Corollary 4.5 (Quadratic convergence rate)**.** *If x_k lies in the local convergence region, then for all $\ell \geq 1$,*

$$\|\nabla f(x_{k+\ell})\| \leq \left(\frac{L}{2m^2}\|\nabla f(x_k)\|\right)^{2^\ell}.$$

Proof. From Theorem 4.4, once $\|\nabla f(x_k)\| < \eta$, we have

$$\|\nabla f(x_{j+1})\| \leq c\|\nabla f(x_j)\|^2, \qquad c := \frac{L}{2m^2}. \tag{4.8}$$

Applying (4.8) repeatedly yields

$$\|\nabla f(x_{k+1})\| \leq c\|\nabla f(x_k)\|^2,$$

$$\|\nabla f(x_{k+2})\| \leq c\|\nabla f(x_{k+1})\|^2 \leq c(c\|\nabla f(x_k)\|^2)^2 = c^3\|\nabla f(x_k)\|^4,$$

and, by induction,

$$\|\nabla f(x_{k+\ell})\| \leq c^{(2^\ell - 1)}\|\nabla f(x_k)\|^{2^\ell}.$$

Since $c\|\nabla f(x_k)\| < 1$ in the local region (by the choice of $\eta < 2m^2/L$), we may write

$$c^{(2^\ell - 1)}\|\nabla f(x_k)\|^{2^\ell} = (c\|\nabla f(x_k)\|)^{2^\ell} \cdot c^{-1}.$$

Absorbing the fixed factor c^{-1} into the bound gives

$$\|\nabla f(x_{k+\ell})\| \leq (c\|\nabla f(x_k)\|)^{2^\ell},$$

which proves the stated quadratic (doubly exponential) rate. □

Theorem 4.6 (Complete convergence result)**.** *Under Assumption 4.2, Newton's method with backtracking line search:*

1. *generates a sequence $\{x_k\} \subset L_0$,*

2. *decreases the objective globally until entering a neighborhood of the minimizer,*

3. *converges quadratically to the unique minimizer $x^\star$.*

Moreover,

$$\lambda(x_k) \to 0, \qquad \frac{\lambda(x_k)^2}{2} \text{ accurately predicts the local objective decrease.}$$

Proof. The proof proceeds in the following steps.

Step 1: Invariance of the level set. By the Armijo condition,

$$f(x_{k+1}) \leq f(x_k),$$

so the sequence $\{f(x_k)\}$ is nonincreasing. Hence

$$x_k \in L_0 \quad \forall k.$$

Since L_0 is closed and bounded (Assumption 4.2), the sequence $\{x_k\}$ is bounded.

Step 2: Global descent phase. By Theorem 4.3, whenever $\|\nabla f(x_k)\| \geq \eta$, the decrease satisfies

$$f(x_{k+1}) - f(x_k) \leq -\gamma,$$

for some fixed $\gamma > 0$.

Since f is bounded below on L_0, only finitely many such decreases can occur. Therefore there exists k_0 such that

$$\|\nabla f(x_k)\| < \eta \quad \forall k \geq k_0.$$

Thus the sequence eventually enters the local convergence region.

Step 3: Local quadratic convergence. For all $k \geq k_0$, Theorem 4.4 applies, yielding

$$\|\nabla f(x_{k+1})\| \leq \frac{L}{2m^2}\|\nabla f(x_k)\|^2.$$

Hence $\|\nabla f(x_k)\| \to 0$ quadratically.

Step 4: Convergence of the iterates. Strong convexity implies the gradient inequality

$$\|\nabla f(x)\| \geq m\|x - x^\star\|.$$

Indeed,

$$\langle \nabla f(x) - \nabla f(x^\star), x - x^\star \rangle \geq m\|x - x^\star\|^2,$$

and since $\nabla f(x^\star) = 0$,

$$\|\nabla f(x)\|\|x - x^\star\| \geq m\|x - x^\star\|^2.$$

Canceling $\|x - x^\star\| \neq 0$ gives

$$\|x - x^\star\| \leq \frac{1}{m}\|\nabla f(x)\|.$$

Thus

$$\|x_k - x^\star\| \leq \frac{1}{m}\|\nabla f(x_k)\| \to 0,$$

so $x_k \to x^\star$.

Uniqueness of the minimizer follows from strong convexity.

Step 5: Behavior of the Newton decrement. By definition,

$$\lambda(x_k)^2 = \nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k).$$

Since

$$\frac{1}{M}\|\nabla f(x_k)\|^2 \leq \lambda(x_k)^2 \leq \frac{1}{m}\|\nabla f(x_k)\|^2,$$

we obtain

$$\lambda(x_k) \to 0.$$

Moreover, in the local region,

$$f(x_k) - f(x_{k+1}) = \frac{1}{2}\lambda(x_k)^2 + \mathcal{O}(\lambda(x_k)^3),$$

so the quadratic model prediction $\lambda(x_k)^2/2$ asymptotically matches the true objective decrease.

□

5 Further Reading

This module develops foundational methods for deterministic and stochastic optimization. The following references provide deeper treatments and modern applications. First-order methods form the backbone of modern deep learning. For optimization perspectives on neural net-

works, see [8, 4]. For large-scale convex and nonconvex first-order theory, see [12, 2]. Adaptive and stochastic gradient methods widely used in practice (e.g., SGD, momentum, Adam) are discussed in [9]. Gradient-based algorithms for matrix optimization, including low-rank factorization and semidefinite problems, are treated in [1, 7]. For convex relaxations and matrix completion, see [6]. Applications to control and signal processing appear in [5].

Classical foundations of stochastic approximation originate in [13]. Comprehensive modern treatments are given in [11, 3]. The ODE method and martingale techniques are central tools in analyzing convergence. Stochastic approximation was originally developed for root-finding problems of the form $h(x) = 0$. Connections to fixed-point iterations and monotone operators are developed in [10]. In reinforcement learning, temporal-difference and Q-learning algorithms can be viewed as stochastic approximation schemes. For foundational treatments, see [14, 16, 15]. These works interpret learning dynamics as noisy approximations to contraction mappings or Bellman fixed points.

6 Exercises

Exercise 1 (Total Variation Image Denoising)**.** In this exercise, we consider total variation denoising for images, which aims to reduce noise while preserving edges.

Let $Y \in \mathbb{R}^{n\times m}$ be a noisy image, where each element of Y is the pixel value of an image of size $n \times m$. From the two-dimensional image data Y, extract each column vector, say $y \in \mathbb{R}^n$, and solve the following optimization problem column by column:

$$\min_{x\in\mathbb{R}^n} \|x - y\|_2^2 + \lambda \sum_{i=1}^{n-1} |x_{i+1} - x_i|. \tag{6.1}$$

The first term measures fidelity to the data, while the second term is the total variation regularization. Develop an algorithm to solve this optimization problem.

Exercise 2 (Branching Process as Stochastic Approximation)**.** Let $\{Z_n\}_{n\geq 0}$ be a Galton–Watson branching process with offspring distribution ξ satisfying $\mathbb{E}[\xi] = m > 0$ and $\mathrm{Var}(\xi) = \sigma^2 < \infty$. The recursion is $Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n,i}$, where $\{\xi_{n,i}\}$ are i.i.d. copies of ξ.

(a) Define $X_n := Z_n/a_n$. Show that if $a_n = m^n$, then $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$, so $\{X_n\}$ is a martingale.

(b) For $Z_n > 0$, define the empirical growth ratio $R_n := Z_{n+1}/Z_n$. Show that $R_n = m + \frac{1}{Z_n}\sum_{i=1}^{Z_n}(\xi_{n,i} - m)$. Hence $R_n = m + \varepsilon_n$ with $\mathbb{E}[\varepsilon_n \mid \mathcal{F}_n] = 0$ and $\mathrm{Var}(\varepsilon_n \mid \mathcal{F}_n) = \sigma^2/Z_n$.

Interpret this as a stochastic approximation with vanishing noise.

(c) Suppose $m > 1$ (supercritical case). Since $Z_n \to \infty$ with positive probability, the noise variance $\sigma^2/Z_n \to 0$ along surviving trajectories. Explain why this suggests $R_n \to m$ almost surely on the survival event.

(d) Define $x_{n+1} = x_n + \frac{1}{n+1}\big(R_n - x_n\big)$ for $Z_n > 0$. Show that this can be written as $x_{n+1} = x_n + \alpha_n\big(h(x_n) + M_{n+1}\big)$ with $\alpha_n = 1/(n+1)$ and $h(x) = m - x$. Identify the limiting ODE $\dot{x} = m - x$ and show that $x_n \to m$ almost surely on the survival event.

(e) Explain why extinction ($Z_n \to 0$) violates the vanishing-noise condition required in stochastic approximation, and why conditioning on survival restores convergence.

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